

Environment-Aware Robust Heterogeneous Federated Learning under Corruption–Label Entanglement

Anonymous submission

Abstract

Model-heterogeneous federated learning becomes especially brittle when visual corruptions are not independent of semantic labels: each client can observe a different, class-dependent corruption pattern, so minimizing average local risk may reinforce client-specific shortcuts and harm tail groups. We study this setting under a controlled corruption–label entanglement protocol with four heterogeneous client architectures, label-skewed private CIFAR-10, unlabeled public CIFAR-100, and strict separation of fit, audit, and final-test roles. Our method adds three local components to a matched AugMix/JSD/DCL base: a calibrated Public Environment Witness (PEW) learned from synthetic corruptions of public images, Balanced Environment Risk (BER) that reweights predicted environments within each observed class, and CDep, a normalized within-class linear dependence surrogate between task features and PEW embeddings. The candidate and control use the same audit-only asymmetric collaboration, so the matched difference isolates the local PEW/BER/CDep mechanism. On one fixed CLE scenario, three matched training seeds yield last-10-round candidate-minus-control changes of $+5.18 \pm 0.44$ average accuracy, $+3.04 \pm 0.56$ worst-client accuracy, $+9.46 \pm 1.79$ worst class–environment accuracy, and -5.14 ± 0.42 class-wise environment gap. Seen, unseen, and group-tail metrics improve as well. These results establish training-seed stability on the fixed scenario, not cross-scenario generalization or state-of-the-art superiority; the missing comparisons and ablations are explicitly identified.

Introduction

Federated learning (FL) coordinates learning from decentralized data without centralizing raw examples (McMahan et al. 2017). In realistic deployments, however, clients can differ simultaneously in model architecture, semantic class support, and data quality. Model-heterogeneous FL therefore relies on architecture-compatible knowledge transfer, for example through subnetwork sharing, public-data distillation, or class prototypes (Diao, Ding, and Tarokh 2021; Lin et al. 2020; Tan et al. 2022; Zhang et al. 2024). A separate line of work studies robustness to noisy or corrupted clients (Fang and Ye 2022; Fang, Ye, and Yang 2023; Fang, Ye, and Du 2025).

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The intersection is difficult because the local corruptions that degrade one architecture can also contaminate cross-model collaboration.

We focus on a sharper failure mode: *corruption–label entanglement* (CLE). Within a client, semantic classes are disproportionately paired with particular corruption environments. A learner can then exploit the environment as a class shortcut even when standard corruption augmentation improves average robustness. Label skew further makes the set and frequency of class–environment cells client-specific. Consequently, average client accuracy can conceal severe failures in rare cells, and a collaboration rule can transfer models that score well on dominant cells while remaining fragile elsewhere.

Figure 1 summarizes both the setting and our intervention. We use an unlabeled public dataset only to train a small environment witness with synthetic environment supervision. On private fit data, witness predictions provide coarse environment groups and continuous environment embeddings without exposing true private corruption operator identities. Balanced Environment Risk (BER) balances predicted environments *within each class*, while CDep penalizes normalized linear dependence between task representations and environment embeddings *within each class*. Both components preserve client-private counts and features. The control and candidate retain the same AugMix/Jensen–Shannon divergence (JSD)/diversity-enhanced contrastive learning (DCL) base (Hendrycks et al. 2020; Fang, Ye, and Du 2025) and the same strict audit-only asymmetric heterogeneous collaboration. Therefore the A/B difference is the local calibrated PEW + BER + CDep mechanism, not the collaboration rule.

Our contributions are intentionally scoped. First, we formalize a controlled CLE-HFL v2 protocol combining four heterogeneous models, Dirichlet label skew, client-specific class–corruption association, seen/unseen corruption operators, and strict fit/audit/final-test roles. Second, we present a local environment-aware mechanism that needs neither private operator metadata nor final-test feedback. Third, under one frozen scenario and three matched training seeds, we report durable 40-round gains in average, worst-client, worst class–environment, class-wise gap, seen/unseen, and group-tail metrics. We do *not* infer cross-scenario generalization, universal robustness, statistical significance, or state-of-the-art superiority from this evidence.

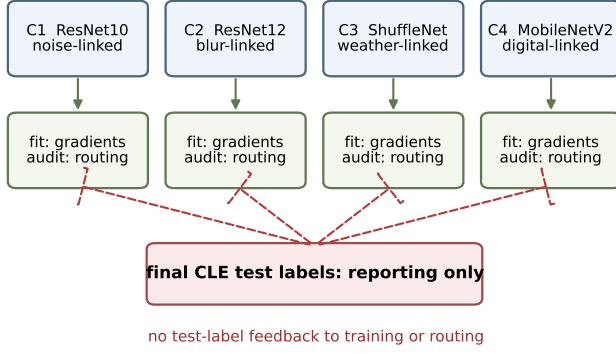
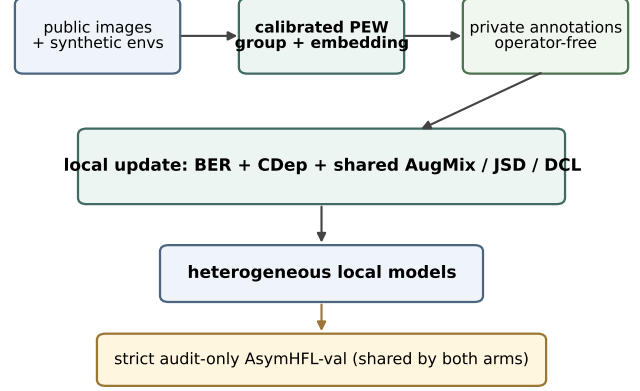
(a) CLE-HFL setting and strict data roles**(b) Our method and matched collaboration control**

Figure 1: (a) Four heterogeneous clients have label-skewed, class-dependent corruption patterns. Fit samples drive local gradients; client-private audit samples are used only for strict teacher routing; final-test labels are reporting-only. (b) Public synthetic supervision trains and calibrates PEW. Private PEW predictions feed BER and CDep, alongside the shared robust base. Strict audit-only AsymHFL-val is shared by candidate and control and is not a causal contribution.

Related Work

Model-heterogeneous FL. HeteroFL trains nested subnetworks and produces a single global inference model, requiring structural compatibility across widths (Diao, Ding, and Tarokh 2021). FedDF instead distills an ensemble of heterogeneous client models on unlabeled server data (Lin et al. 2020). FedProto communicates class prototypes rather than parameters, while FedTGP learns trainable global prototypes with adaptive-margin contrastive learning (Tan et al. 2022; Zhang et al. 2024). These methods address model and/or statistical heterogeneity, but they do not directly target class-dependent corruption environments. Our protocol permits unrelated client architectures and already uses a public-data collaboration base; the proposed difference is local environment-aware risk shaping.

Robust heterogeneous FL. RHFL considers label-noisy heterogeneous clients and aligns feedback on public data using confidence reweighting (Fang and Ye 2022). AugHFL studies common data corruptions and combines robust local augmentation with reweighted heterogeneous communication (Fang, Ye, and Yang 2023). RAHFL extends this direction with DCL and asymmetric one-way collaboration that avoids lower-quality teachers (Fang, Ye, and Du 2025). Federated robustness propagation addresses adversarial, rather than common-corruption, robustness using batch-normalization statistics (Hong et al. 2023). Our matched control inherits the strong local augmentation/DCL and asymmetric collaboration design. CLE-HFL differs in making corruption exposure class-dependent, which motivates balancing and dependence control conditioned on class rather than only client-level quality.

Corruption and group robustness. Common-corruption benchmarks expose large robustness gaps not captured by clean accuracy (Hendrycks and Dietterich 2019); AugMix

improves corruption robustness through stochastic augmentation chains and a JSD consistency objective (Hendrycks et al. 2020). Group DRO optimizes worst-group risk when group identities are supplied (Sagawa et al. 2020). Our predicted environment groups are imperfect and are available only locally; BER uses them for a capped, class-conditional reweighting target rather than claiming a worst-group DRO guarantee. CDep further discourages within-class linear association between task features and continuous PEW embeddings. This is a dependence surrogate, not evidence of conditional independence.

A primary-source audit accompanying the source release records each cited method’s architecture assumptions, public-data/logit requirements, robustness target, and comparability. Because no external method has yet been rerun under the exact frozen CLE split and strict data roles, we avoid a state-of-the-art claim.

Problem Formulation and Strict Protocol

Let client $k \in \{1, \dots, K\}$ own a heterogeneous classifier $f_k = h_k \circ g_k$ with private labeled examples $D_k = \{(x_i, y_i)\}$. Architectures and representation dimensions may differ. Each example is generated under a latent corruption operator and severity, but these private operator identifiers are unavailable to optimization. CLE means that the environment distribution depends on both client and class,

$$p_k(E | Y = c) \neq p_k(E), \quad (1)$$

and can be highly concentrated. Label skew changes $p_k(Y)$ and can remove classes from a client entirely, creating sparse class–environment support.

Strict data roles. Each client’s private data is persistently split in a class-stratified 85/15 ratio into D_k^{fit} and D_k^{audit} .

Only D_k^{fit} contributes local gradients or PEW-derived private annotations. The audit subset remains client-private and is used solely to order or route teachers in the shared asymmetric collaboration. Final CLE test labels are used only to produce reporting metrics after training; they do not enter gradients, routing, model selection, early stopping, or hyperparameter tuning. The public CIFAR-100 labels are not consumed: public images receive only synthetic environment supervision for PEW.

Metrics. For client accuracy A_k , $\text{Avg} = K^{-1} \sum_k A_k$ and $\text{Worst} = \min_k A_k$. Let $A_{c,e}$ denote accuracy in each evaluator-valid class-environment cell aggregated over its defined test domain. Worst Class-Environment Accuracy is

$$\text{WCCA} = \min_{(c,e) \in \mathcal{V}} A_{c,e}, \quad (2)$$

where $\mathcal{V}$ contains cells with evaluator-valid support. Class-wise Fairness Gap averages, over valid classes, the range across valid environments,

$$\text{CFG} = \frac{1}{|\mathcal{C}_{\mathcal{V}}|} \sum_{c \in \mathcal{C}_{\mathcal{V}}} \left(\max_{e \in \mathcal{V}_c} A_{c,e} - \min_{e \in \mathcal{V}_c} A_{c,e} \right). \quad (3)$$

Thus higher Avg, Worst, and WCCA are better, whereas lower CFG is better. We also report seen/unseen operator averages and worst values, worst corruption-group accuracy, and worst client-group accuracy.

Method

Calibrated Public Environment Witness

PEW is a small convolutional network trained on a random subset of 5,000 unlabeled public CIFAR-100 training images. Synthetic transforms provide an environment target from

$$\mathcal{E} = \{\text{clean, noise, blur, weather, digital, unknown}\} \quad (4)$$

and a non-clean severity target in $\{1, \dots, 5\}$. The network outputs environment logits $q_\phi(x)$, severity logits $r_\phi(x)$, and an embedding $u_\phi(x) \in \mathbb{R}^{32}$. The verified training objective is

$$\mathcal{L}_{\text{PEW}} = \text{CE}(q_\phi(x), e) + 0.25 \mathbf{1}[s > 0] \text{CE}(r_\phi(x), s - 1). \quad (5)$$

We train for five epochs with Adam at learning rate 10^{-3} and restore the checkpoint with highest public-validation environment accuracy. On a disjoint 20% public validation split, a confidence threshold $\tau \in \{0, .01, \dots, .95\}$ is selected to maximize environment accuracy after replacing predictions with confidence below τ by *unknown*; ties choose the smaller threshold. PEW is then frozen. For each private fit example, it supplies $\hat{e}_i$, confidence, and u_i ; it never receives private class labels or true private operator metadata during PEW training.

Balanced Environment Risk

For client k , observed class c , and predicted environment e , let

$$n_{k,c,e} = |\{i \in D_k^{\text{fit}} : y_i = c, \hat{e}_i = e\}|. \quad (6)$$

Only groups with $n_{k,c,e} \geq 2$ are valid. With support exponent $\gamma_s = 0.5$ and cap $n_{\text{cap}} = 32$, the within-class target mass is

$$a_{k,c,e} = \frac{\min(n_{k,c,e}, n_{\text{cap}})^{\gamma_s}}{\sum_{e' \in \mathcal{V}_{k,c}} \min(n_{k,c,e'}, n_{\text{cap}})^{\gamma_s}}. \quad (7)$$

Each valid example receives

$$w_{k,i} = \frac{1}{|C_k^{\text{valid}}|} \frac{a_{k,y_i,\hat{e}_i}}{n_{k,y_i,\hat{e}_i}}. \quad (8)$$

For a uniformly sampled minibatch B from $N_k = |D_k^{\text{fit}}|$ examples,

$$\hat{\mathcal{L}}_{\text{BER}} = \frac{N_k}{|B|} \sum_{i \in B} w_{k,i} \text{CE}(f_k(x_i), y_i). \quad (9)$$

This scaling makes the minibatch estimate consistent with the client-level weighted objective. BER replaces ordinary clean cross-entropy; it is not a second classification loss. Capping prevents dominant cells from absorbing all mass, while $\gamma_s > 0$ avoids assigning equal target mass to cells with negligible support. All counts stay local.

Conditional Environment Dependence Penalty

Let $z_i = g_k(x_i)$ and use a fixed seeded random projection P_k to obtain $z_i^p = z_i P_k \in \mathbb{R}^{64}$. Projection columns are normalized and scaled; P_k is not trained. For each minibatch class with at least three samples, task projections and PEW embeddings are centered and standardized dimension-wise. Their empirical cross-covariance is $C_{k,c}$. We penalize

$$\mathcal{L}_{\text{CDep}} = \frac{1}{|C_{k,B}^{\text{valid}}|} \sum_{c \in C_{k,B}^{\text{valid}}} \text{mean}(C_{k,c}^{\odot 2}). \quad (10)$$

PEW embeddings are detached, so CDep updates only the task model. Class conditioning is essential: an unconditional penalty could suppress legitimate class signal whenever class and environment are entangled. The normalized squared cross-covariance detects linear dependence in the projected representation only; it neither estimates mutual information nor proves conditional independence.

Local Objective and Collaboration

For an AugMix clean view and two stochastic mixtures, the implementation uses the shared JSD consistency term and DCL base from the matched control. The validated candidate objective is

$$\mathcal{L}_k = \hat{\mathcal{L}}_{\text{BER}} + 12\mathcal{L}_{\text{JSD}} + \mathcal{L}_{\text{DCL}} + 0.05\mathcal{L}_{\text{CDep}}. \quad (11)$$

No EBST or SCP term is active. After local updates, both arms use the identical strict AsymHFL-val collaboration, whose teacher ordering is computed only on D_k^{audit} . This collaboration can affect both arms but cannot explain their matched difference.

Complexity. PEW is trained once and frozen; its private inference produces cached annotations. BER adds local count bookkeeping and scalar weights. CDep adds a fixed 64-dimensional projection and class-wise covariance computation per minibatch. Exact wall-clock, memory, and amortized PEW costs remain a pre-submission measurement TODO rather than an estimated claim.

Table 1: Frozen CLE-HFL v2 protocol.

Item	Setting
Private/public data	CIFAR-10 / CIFAR-100
Clients / models	4 / four distinct CNNs
Label skew / CLE	$\alpha = 0.5 / \gamma = 0.9$
Scenario	seed0_split0
Operators	11 seen + 4 unseen
Train seeds / rounds	{0, 1, 2} / 40
Local epochs	1
Public batches/round	4
Fit/audit	class-stratified 85/15
Final-test labels	reporting only

Experiments

Setup and Matched Comparison

Table 1 gives the frozen protocol. Private CIFAR-10 is partitioned across four clients with Dirichlet $\alpha = 0.5$ and CLE strength $\gamma = 0.9$. Client models are ResNet10, ResNet12, ShuffleNet, and MobileNetV2. The corruption set contains 11 seen and 4 unseen operators. All runs use 40 rounds, one local epoch, four public batches per round, and training seeds 0, 1, and 2. These seeds alter matched initialization and training randomness only; the client class partitions, operator map, and fit/audit split remain the same `seed0_split0` scenario. Therefore the experiment tests training-seed stability on one scenario.

The control is AugMix + JSD + DCL + strict AsymHFL-val. The candidate adds calibrated PEW, replaces clean CE with BER, and adds CDep while keeping the collaboration fixed. Data, architecture assignment, optimizer, seed, local epochs, public batches, evaluation cadence, and split are matched.

Table 2 reports last-10-round means. Every seed improves all four primary metrics in the desired direction. Across the three training seeds, the mean changes are +5.18 points in Avg, +3.04 in Worst, +9.46 in WCCA, and -5.14 in CFG. Figure 2 visualizes dispersion across the three matched seeds; error bars are sample standard deviations, not confidence intervals. The archived 12-round attribution check also passes for every seed, with mean last-five changes of +4.59 Avg, +4.22 Worst, +5.55 WCCA, and -6.72 CFG. Thus the 40-round result is not merely a late-window artifact relative to that earlier checkpoint.

Durability and Tail Analysis

The frozen durability gates require last-10 deltas of at least +1.5 Avg, +1.0 Worst, nonnegative WCCA, and at most -1.0 CFG, plus same-direction last-five anti-collapse gates. All eight gates pass for each of three seeds (24/24). Each arm contains exactly rounds 0–39, no core metric is missing, and the first 12 rounds reproduce the earlier matched run according to the supplied experiment audit.

Table 3 shows that gains extend beyond client averages. Mean worst corruption-group and worst client-group accuracy improve by 7.04 and 6.36 points. Seen and unseen operator averages improve by 5.06 and 5.51 points, while their

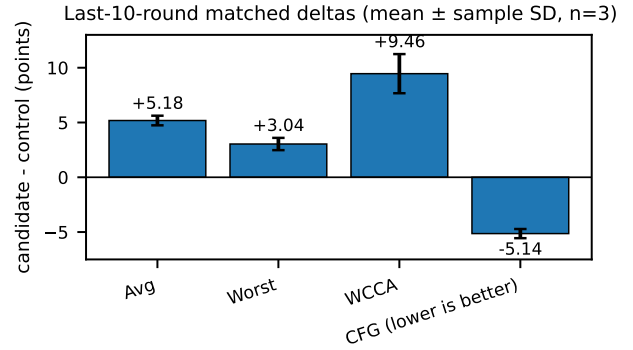


Figure 2: Last-10-round matched deltas over three training seeds on the same fixed scenario. Error bars show sample SD; $n = 3$ does not support a significance claim.

worst values improve by 2.91 and 3.38 points. The unseen result is evaluation on operators held outside the PEW seen-operator set under the fixed scenario; it is not evidence of arbitrary real-world or independently regenerated scenario generalization.

Evidence Still Required

The current matched control is scientifically useful for causal attribution, but it cannot establish superiority to the broader literature. In particular, prototype, distillation, subnetwork, and robust-HFL baselines differ in architecture compatibility, public-data access, and collaboration protocol. Comparable methods must be adapted and rerun under the exact strict data roles. Likewise, component ablations are necessary to determine whether BER and CDep are individually beneficial and whether PEW calibration is essential. We keep these cells as explicit TODOs instead of imputing numbers.

Limitations, Ethics, and Reproducibility

Evidence boundary. The principal limitation is scenario count: three training seeds share one frozen class partition, operator assignment, and fit/audit split. The reported sample variability therefore measures optimizer and initialization stability, not uncertainty over CLE scenarios. The public/private dataset pair is also narrow, and current experiments do not cover non-visual modalities, larger models, partial participation, privacy attacks, or public-data mismatch. PEW groups are hand-defined coarse families and synthetic supervision can misrepresent deployment corruptions. BER depends on predicted groups; systematic PEW errors can create incorrect balancing. CDep controls only normalized linear cross-covariance after a fixed projection and may leave nonlinear environment information intact.

Privacy and information use. The method keeps raw private data, class–environment counts, features, and PEW annotations on clients. This is data locality, not a formal privacy guarantee: embeddings, logits, or model updates may still leak information under known attacks. Differential privacy, secure aggregation, and leakage audits are not evaluated. The

Table 2: Matched 40-round comparison on the fixed `seed0_split0` CLE scenario. Entries are last-10-round means in percentage points. The last four columns are candidate minus control; lower CFG is better. The mean $\pm$ SD row summarizes the three matched training seeds and is not a significance test.

Seed	Control				Calibrated PEW + BER + CDep				Δ			
	Avg	Worst	WCCA	CFG	Avg	Worst	WCCA	CFG	Avg	Worst	WCCA	CFG
0	40.15	31.49	4.45	30.50	45.08	34.79	14.33	25.03	+4.93	+3.30	+9.88	-5.47
1	39.99	31.48	4.18	30.82	45.68	34.90	11.68	25.53	+5.69	+3.42	+7.50	-5.29
2	40.10	31.49	3.90	30.09	45.03	33.89	14.90	25.42	+4.93	+2.39	+11.00	-4.67
Mean $\pm$ SD	–	–	–	–	–	–	–	–	5.18 $\pm$ 0.44	3.04 $\pm$ 0.56	9.46 $\pm$ 1.79	-5.14 $\pm$ 0.42

Table 3: Last-10-round candidate-minus-control tail and operator-split gains on the fixed scenario (percentage points).

Metric	Seed 0	Seed 1	Seed 2	Mean
Worst group	6.43	8.20	6.49	7.04
Worst client-group	6.09	6.90	6.09	6.36
Seen Avg	4.75	5.73	4.70	5.06
Seen Worst	3.20	3.30	2.24	2.91
Unseen Avg	5.43	5.57	5.54	5.51
Unseen Worst	3.56	3.75	2.83	3.38

Table 4: Evidence deliberately left open rather than fabricated.

Question	Status
External matched baselines	TODO: rerun comparable HFL methods under the identical CLE split and strict data roles.
Component attribution	TODO: PEW-only, BER-only, CDep-only, and pairwise ablations.
Cross-scenario robustness	TODO: regenerate class partitions and operator maps with independent scenario seeds.
Efficiency	TODO: wall-clock, memory, communication, and PEW amortization measurements.
Statistical inference	TODO: more independent scenario seeds; $n = 3$ training seeds alone is insufficient.

client-private audit subset affects teacher routing, so it is validation information even though it is excluded from gradients. Final-test labels are reporting-only. These distinctions should be retained in any implementation or comparison.

Reproducibility. The source package records the exact scenario, seed semantics, fit/audit hash, formulas, hyperparameters, result tables, and primary-source literature audit. The validated code head is recorded in the build report, but the anonymous manuscript does not link to a potentially identifying repository. Before submission, authors should provide an anonymous code/data artifact if permitted, complete the separate AAAI reproducibility checklist, add hardware/runtime measurements, and release raw per-round logs needed to reproduce curves and uncertainty analyses.

Potential impact. More robust decentralized models may reduce failures for underrepresented class-environment groups, but a coarse environment taxonomy can also conceal deployment-specific harms. Tail metrics should be complemented with domain-relevant safety evaluation, and audit sets should not be repurposed in ways that blur the separation from final testing.

Conclusion

We studied heterogeneous federated learning when client-specific corruptions are entangled with labels. Calibrated PEW supplies private environment predictions from public synthetic supervision, BER balances predicted environments within class, and CDep discourages within-class linear dependence between task and environment representations. Under a strict matched protocol, the mechanism improves average, worst-client, class-environment tail, fairness-gap, seen/unseen, and group-tail metrics for three training seeds over 40 rounds on one fixed CLE scenario. The result supports durable fixed-scenario training-seed stability. Cross-scenario evaluation, matched external baselines, component ablations, efficiency measurements, and stronger statistical inference remain necessary before broader or state-of-the-art claims.

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